



(a) Order Urodela.
Urodeles (salamanders) retain their tail as adults.



(b) Order Anura.
Anurans (toads and frogs) lack a tail as adults.



(c) Order Apoda.
Apodans, or caecilians, are legless, mainly burrowing amphibians.

▲ Figure 34.22 Amphibians.

also found in early terrestrial tetrapods (Figure 34.22a). Paedomorphosis is common among aquatic salamanders; the axolotl, for instance, retains larval features even when sexually mature (see Figure 25.25).

Frogs

Numbering about 5,420 species, anurans, or frogs, are better suited than salamanders to locomotion on land (Figure 34.22b). Adult frogs use their powerful hind legs to hop along the terrain. Although often distinctive in appearance, the animals known as “toads” are simply frogs that have leathery skin or other adaptations for life on land. A frog nabs insects and other prey by flicking out its long, sticky tongue, which is attached to the front of the mouth. Frogs display a great variety of adaptations that help them avoid being eaten by larger predators. Their skin glands secrete distasteful or even poisonous mucus. Many poisonous species have color patterns that camouflage them or have bright coloration, which predators appear to associate with danger (see Figure 54.6).

Caecilians

The approximately 170 species of apodans, or caecilians, are legless and nearly blind, and superficially they resemble earthworms (Figure 34.22c). Their absence of legs is a secondary adaptation, as they evolved from a legged ancestor. Caecilians inhabit tropical areas, where most species burrow in moist forest soil.



(a) The tadpole is an aquatic herbivore with a fishlike tail and internal gills.



(b) During metamorphosis, the gills and tail are resorbed, and walking legs develop. The adult frog will live on land.

(c) The adults return to water to mate. The male grasps the female, stimulating her to release eggs. The eggs are laid and fertilized in water. They have a jelly coat but lack a shell and would desiccate in air.



▲ Figure 34.23 The “dual life” of a frog (*Rana temporaria*).

Lifestyle and Ecology of Amphibians

The term *amphibian* (derived from *amphibious*, meaning “both ways of life”) refers to the life stages of many frog species that live first in water and then on land (Figure 34.23). The larval stage of a frog, called a tadpole, is usually an aquatic herbivore with gills, a lateral line system resembling that of aquatic vertebrates, and a long, finned tail. The tadpole initially lacks legs; it swims by undulating its tail. During the metamorphosis that leads to the “second life,” the tadpole develops legs, lungs, a pair of external eardrums, and a digestive system adapted to a carnivorous diet. At the same time, the gills disappear; the lateral line system also disappears in most species. The young frog crawls onto shore and becomes a terrestrial hunter. In spite of their name, however, many amphibians do not live a dual—aquatic and terrestrial—life. There are some strictly aquatic or strictly terrestrial frogs, salamanders, and caecilians. Moreover, salamander and caecilian larvae look much like the adults, and typically both the larvae and the adults are carnivorous.

Most amphibians are found in damp habitats such as swamps and rain forests. Even those adapted to drier habitats spend much of their time in burrows or under moist leaves, where humidity is high. One reason amphibians require relatively wet habitats is that they rely heavily on their moist skin for gas exchange—if their skin dries out, they cannot get enough oxygen. In addition, amphibians typically lay their eggs in water or in moist environments on land; their eggs lack a shell and dehydrate quickly in dry air.

Fertilization is external in most amphibians; the male grasps the female and spills his sperm over the eggs as the female sheds them (see Figure 34.23c). Some amphibian species lay vast numbers of eggs in temporary pools, and egg mortality is high. In contrast, other species lay relatively few